Research Publications, Preprints, and Upcoming Papers

Emiliano Rizza

Selected research contributions

Research Publications

Lorentz symmetry violating Lifshitz-type field theories

E. Rizza, D. Zappalà

Modern Physics Letters A **37**, No. 30, 2250203 (2022)

11 journal pages; 13 pages in the arXiv version

Journal / DOI | arXiv:2209.11060

Contribution. My contribution focused on the perturbative quantum-field-theory analysis extending the earlier scalar-only Lifshitz framework to systems containing fermionic degrees of freedom and to the consistency issues associated with gauge fields. I worked on the ultraviolet power counting and loop corrections, including scalar and fermionic tadpoles and fermion-loop contributions, and on the renormalization of the infinite families of scalar and Yukawa-like couplings, showing how the $z = 3$ higher-derivative propagators soften ultraviolet divergences to logarithmic behavior, how the Yukawa sector becomes asymptotically free, and how stability conditions enter the running of the scalar couplings. I also contributed to the analysis of the quantum corrections to the scalar mass and their impact on the hierarchy problem.

Related earlier work: D. Zappalà, *Ultraviolet properties of Lifshitz-type scalar field theories*, arXiv:2111.08385.

Preprints

The quadratic growth of Krylov spread complexity in the BTZ black hole

A. Bhattacharya, M. Flory, M. P. Heller, **E. Rizza**, T. Schuhmann

submitted to PRX journal

37 pages (22 pages main text + 15 pages appendices)

arXiv:2608.09922

Contribution. My contribution centered on the bulk side of the project, from the geometric construction to its analytical and numerical implementation, while also supporting the validation and revision of the manuscript as a whole. I developed and tested the generalized bulk observable on maximal-volume slices and its early-time expansion, including the order-by-order matching of an infinite tower of extrinsic-curvature invariants to the boundary Krylov series. The bulk-boundary matching was combined with large-order fits and analytic resummation of the coefficient tail, allowing us to identify the singular final-slice behavior needed to evade the generic linear-growth regime of finite Complexity=Anything functionals and reproduce the quadratic behavior suggested by the boundary continuation. The project therefore required extensive Mathematica-based workflows with the analytic construction of a candidate geometric dual of Krylov spread complexity in AdS_3/CFT_2.

Upcoming Papers and Ongoing Research

Rényi second laws for Charged AdS Black Hole

A. Bernamonti, F. Galli, **E. Rizza**

Manuscript in preparation; current draft: 43 pages

Public link: forthcoming

Contribution. My contribution develops the charged extension of the Rényi second-law programme to Reissner–Nordström–AdS_4 black holes. I worked on formulating and comparing information-theoretic constraints formulated in terms of the monotonicity properties of Rényi entropies in order to set more stringent bounds than those imposed by the area increase theorem. The analysis covers mergers with equal, different, and opposite charges, including extremal configurations, and derives constraints on the final mass and temperature while accounting for the charged AdS phase structure. A complementary part of the project studies Hawking–Rényi monotones and the flat-space limit, and connects these information-theoretic bounds with effective merger dynamics and radiative-efficiency estimates through a combination of analytical approximations and numerical analysis of gravitational and electromagnetic

emission.

Extension of: A. Bernamonti, F. Galli, R. C. Myers, I. A. Reyes, *Rényi second laws for black holes*, JHEP 10 (2024) 177; arXiv:2407.01753.

Analytical reconstruction of Krylov complexity from partition functions: Edgeworth and large-deviation methods

A. Bhattacharya, M. Flory, M. P. Heller, **E. Rizza**, T. Schuhmann
In preparation / to appear; page count to be determined
Public link: forthcoming

> **Contribution.** My contribution focuses on developing a complementary analytical route for reconstructing Krylov dynamics directly from thermal partition-function data, without first generating a long finite Lanczos chain. The two descriptions used probe complementary regimes: the Edgeworth formulation reconstructs perturbatively the associated orthogonal polynomials and the Lanczos coefficients, while the large-deviation formulation resums it into global continuum data. This provides an analytical counterpart to the high-order numerical reconstruction developed in the previous project on Krylov complexity in BTZ black holes and a route toward systematic applications in high-temperature and large-central-charge CFT_d.

Krylov complexity, orthogonal polynomials and the ASEP model

E. Rizza, M. Watanabe *(collaboration in preparation; author list provisional)*
Upcoming project / manuscript in preparation; page count to be determined
Public link: forthcoming

> **Contribution.** My role in this collaboration is to connect the Krylov-complexity framework with recurrence structures arising in exactly and stochastically solvable models. Building on Watanabe's JT/KPZ correspondence, the project aims to explore how the Schwarzian limit of double-scaled SYK is related to the weakly asymmetric limit of open ASEP. This makes it possible to translate dynamical questions about spreading in Krylov space into questions about recurrence coefficients, spectral edges, and universality classes. The project is designed to build a bridge between holographic complexity, spectral theory, orthogonal-polynomial methods, and nonequilibrium stochastic dynamics.

Related work: M. Watanabe, *A JT/KPZ correspondence*, arXiv:2511.02529.